5 Thematic Synthesis

To Recap: Our study's aim was to explore present day machine learning techniques and in turn to use these techniques to explore the visual nature of London's Built Environment.

Our study's method was twofold: Firstly to enquire whether we could reach as reasonable accuracy rates on a CNN Image Classification Task on London New Development Building Types as achieved by Kang et Al in their US Cities Study and Secondly to apply Image Embedding methods to further explore our model by re-using the image vectors produced in the earlier CNN Training Stage.

Early runs of the workflow achieved accuracy rates of > 66 % on best of class label sets, and > 75% on One Hot Label Runs. Our strategy for dealing with early signs of model overfitting involved 4 approaches:

1. Secondary Validation of the Address Data
2. Applying common Model Fine Tuning Methods
3. Varying Label Set Creation Approaches
4. Visualizing Classification Decisions made by the CNN

None of the 4 achieved significant improvements upon the initial promising classification results. Consequently we conclude that this must point to specific characteristics of the nature of the environment that is being modeled and examined.

The wide spread prevalence of ambiguous label classifications in our London Data underlines the predominance of a Mixed Use policy in current planning decisions policy. In addition the high levels of occluded and invalid unusable images in the London street level image dataset highlights the predominance of building interfaces which lean to the upper more Private/Inaccessible End of the scale in Dovey and Woods nomenclature.

The difficulties which our study came across can also be seen to emphasize the challenges faced in applying Image Classification Machine Learning techniques to the complexity and ambiguity of a modern urban environment(see the recent reassessment of delivery times for robust driverless cat technology). In effect the lessons of applying AI to real world situations underlines the differences between Elegant AI and the reality of complex hybrid pragmatic AI implementations that illustrates the state of the science as at odds with the current media and political narrative framing.

Potential beneficial applications for the analysis and workflow methods explored can be found in the OSM Missing Maps and Facebook Map Challenge using AI to assist in Mapping and indexing the vast amounts of uncharted geo information, or at least as a powerful tool for aiding the workflow and management of the vast geo informational task at hand.

These methods also allude to the potential and feasibility of fresh thinking in terms of Aggregate Design solutions that if realized allow us to re-appraise and re-approach architecture, such as those espoused in Christopher Alexander's Synthesis of Form and the Timeless Way of Building. Here an approach to Urban Design and Architecture is proposed which re-considers aesthetics on a scale that is both local and global and levels.

One further outcome of the study is the material it allows us to apply to a Cadastral narrative on real time management and optimization of a societies building stock.

Observing that London's urban environment is changing is in no way surprising. However taken in context with our study's cognitive and AI exploratory framework there are two key points that we think might be worth highlighting. Firstly, that machine learning methods applied to aggregate data models can provide us with a novel way of measuring, visualizing and ultimately measuring the rates and degrees of change that we are faced with. Future Tools in Urban Design and Planning process and in the new media communication of this process may well look a bit like the tools we have touched upon so far.

Secondly, our exploration of AI as an Image Classification task applied to Urban Visual Artifacts has brought us into contact with the core idea of Visual Semantics. Earlier we highlighted the role of Vision in the development of Cognitive Science, and the milestone of Gibson's Ecological Phenomenological approach in the progress toward embodied and representational theories of AI.

Do the strong taxonomic ambiguities that our CNN Analysis workflow encountered point to a quantifiable characteristic of the cognitive urban environment? Is a change to the Visual Grammar of our conditioning and developmental environment accompanied by change, in a Cognitive sense, on an equally fundamental and intrinsic Level?

These are questions that are presently being addressed in multi disciplinary fields such as the Leverhulme Doctoral Training Programme for the Ecological Study of the Brain which combines psychology, neuroscience, architecture, computer science, engineering, education and geography to measure behavior and brain activity in the Urban environment.

6 Conclusion

Our study's results, whilst not benefiting from the opportunity to fine tune and up scale our model using the most currently effective architectures(e.g. ResNet), were close but did not match the accuracy of the Kang et al building instance study on US and Canadian cities(>70%).

Possible reasons have been highlighted and may be related to the nature of the differing urban environments and differing socio economic approaches to the basic urban issue of urban density. In the process they open up possibilities regarding the secondary motivation of the study which was to determine whether a stylistic narrative could be effected though the application of Machine Learning techniques on an urban centric image classification exercise.

Further refinement of the adopted workflow would be required to feel confident on this point, however differences have been identified between geo graphic locales, the datasets that describe them and some of practical issues encountered in applying CNNs to real world Image Classification tasks explored.

This study has granted us experience in applying AI to a real world challenge, specifically symbolic neural network based AI. In the study's exploration of building typology it has also allowed us to observe the role of semantics, both visual and conceptual and its impact on using connectivist tools to make sense of a human and urban environment.

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